

Lam Fung Academy

SSPA Mock Exam (3) · Primary 5 Mathematics · Mid-term of next semester

Scope: Lessons 21–28 (Circle, Solid Section, Origami Pattern, Volume, Multiplication and Division of Fractions, Division of Decimals, Comprehensive Application)

Exam time: 50 minutes Full score: 100 points

Name: _____

date: _____

Fraction: _____ / 100

Part A: Multiple choice questions

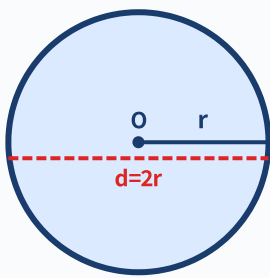
Part A - Multiple Choice Questions

10 questions × 3 points = 30 points

Instructions: Choose the correct answer and fill in the letter of the answer in the space to the right of each question. There is only one correct answer for each question.

Exam tips (SSPA must read)

- ① Do the questions you know first, don't get stuck on one!
- ② Check the unit for each question (cm vs cm² vs cm³)!
- ③ Geometry questions: If there is an elevation in the picture, use the height. If there is no elevation, use the formula to find it!
- ④ Application question: Write an answer sentence! Points will be deducted for not writing steps!
- ⑤ 5 minutes left: Check if the MC has been filled in and if the unit is correct!

question number	Question	fraction
1	▲ Circle: radius 9cm  The radius of a circle is 9 cm. What is the diameter? A. 4.5 cm B. 9 cm C. 18 cm D. 27 cm	___/ 3 points

2	A cylinder cut parallel to the base. What is the shape of the cross section? A. Rectangle B. Triangle C. Circle D. Trapezoid	___/ 3 points
3	Which of the following expanded diagrams cannot be folded into a cube? A. One-four-one type (4 squares in the middle + 1 square above and below) B. Two-three-one type (3 squares in the middle + 2+1 squares on the sides) SEP C. Field shape (2 × 2 squares + 1 upper and lower square) D. Two-two-two ladder type	___/ 3 points
4	▲ Cuboid: 8×5×4cm  The cuboid is 8 cm long, 5 cm wide, and 4 cm high. Volume = ? A. 17 cm³ B. 80 cm³ C. 120 cm³ D. 160 cm³	___/ 3 points
5	calculate $\frac{3}{4} \times \frac{2}{5} = ?$ A. $\frac{5}{9}$ B. $\frac{6}{20}$ C. $\frac{3}{10}$ D. $\frac{6}{5}$	___/ 3 points
6	In the unfolded view of a cube, what are the characteristics of the face opposite the specified face? A. Must be adjacent B. Must be separated by at least one face C. Must be at the diagonal position of the expanded diagram D. There is no pattern	___/ 3 points
7	The side length of the cube water tank is 20 cm. Contains $\frac{3}{4}$ water. What is the volume of water in cm ³ ? A. 6000 cm ³ B. 8000 cm ³ C. 2000 cm ³ D. 3000 cm ³	___/ 3 points
8	Which of the following statements about circles is incorrect SEP ? A. The diameter is the longest segment in the circle B. The length of the diameter is twice the radius C. The length of all radii is equal D. The radius is twice the diameter D. The radius is twice the diameter	___/ 3 points

question number	Question	fraction
9	The length, width, and height of a cuboid are each magnified by 2 times. How many times the new volume is the original? A. 2 times B. 4 times C. 6 times D. 8 times	___ / 3 points
10	In the formula for the area of a trapezoid, what does height mean? A. The length of the hypotenuse of the trapezoid B. The vertical distance between the upper and lower bases C. The length of the upper base D. The length of the lower base	___ / 3 points

Part B: Short answer questions

Part B - Short Questions/Calculation Questions

8 questions × 5 points = 40 points

Instructions: Write the complete calculation steps and answers in the answer area below each question. Answers must be reduced (if applicable) and given in correct units.

question number	Question	fraction
1	The diameter of a circle is 34 cm. Find the length of the radius. _____	___ / 5 points
2	A cuboid is 12 cm long, 7 cm wide, and 5 cm high. (a) Find the area of the base. (2 points) (b) Find the volume. (3 points) _____	___ / 5 points
3	Calculation: $\frac{2}{3} \times \frac{5}{6} = ?$ Write the steps for multiplication and reduction. _____	___ / 5 points
4	Calculation: $\frac{3}{4} \div \frac{1}{2} = ?$ Write down the steps for calculation (multiplying reciprocals). _____	___ / 5 points

5	A cone is cut in a direction passing through the vertex and perpendicular to the base. What is the shape of the cross section? Draw diagram annotations.	____ / 5 points
---	---	------------------------------------

question number	Question	fraction
6	The side length of a cube is $1\frac{1}{2}$ cm. Find the volume. (Answer with fractions or mixed numbers)	___ / 5 points
7	A rectangular water tank has an internal length of 40 cm, a width of 25 cm, and a height of 30 cm. (a) What is the volume of the water tank in cm^3 ? (2 points) (b) How many L of water can the water tank hold? (1 L = 1000 cm^3) (1 point) (c) If $\frac{2}{5}$ is filled with water, how many L of water is there? (2 points)	___ / 5 points
8	A trapezoid (upper base 5 cm, lower base 11 cm, height 6 cm) is the base of a cylinder. The cylinder height is 10 cm. Find the volume of the cylinder. Tips: First find the area of the trapezoid (the area of the base), and then multiply it by the height of the cylinder.	___ / 5 points

Part C - Application Problems3 questions $\times$ 10 points = 30 points

Instructions: Each question must list all calculation steps and write a complete answer. The answer must be accompanied by the unit. The calculation process accounts for steps.

Question 1 (10 points) – Comprehensive application of volume

___ / 10 points

A rectangular swimming pool is 25 m long, 10 m wide, and 2 m deep.

- (a) What is the volume of the swimming pool in m^3 ? (2 points)
- (b) If the water depth is only $\frac{4}{5}$, what is the volume of water m^3 ? (3 points)
- (c) If 5 m^3 of water is injected per minute, how many minutes will it take to fill the empty tank to the level in (b)? (3 points)
- (d) The bottom and four walls of the swimming pool need to be tiled. What is the total area in m^2 to be tiled? (2 points)

Tip: (d) Base area + area of four walls (note that the wall has two long walls and two short walls)

Answer area - please write all calculation steps and complete answer sentences:

Question 2 (10 points) – Mixed use of four fractions

___ / 10 points

There are 180 blocks in a box.

in $\frac{1}{5}$ It's red, $\frac{1}{4}$ It's blue, $\frac{1}{3}$ is green and the rest is yellow.

- (a) How many red blocks are there? (2 points)
- (b) How many blue and green blocks are there? (3 points)
- (c) What fraction of the total bricks of three colors (red + blue + green) account for? (2 points)
- (d) How many yellow blocks are there? What fraction of the total? (3 points)

Answer area - please write all calculation steps and complete answer sentences:

Question 3 (10 points) – Cross-topic comprehensive trap question**____ / 10 points**

A cube water tank A with side length 30 cm. Another rectangular water tank B is 50 cm long, 20 cm wide, and 25 cm high.

- (a) Find the volume of tank A. (Answer in cm^3) (2 points)
- (b) Find the volume of tank B. (Answer in cm^3) (2 points)
- (c) After tank A is filled with water, pour all the water into the empty tank B. How high will the water level be in box B? Will the water overflow? (3 points)
- (d) If poured from box B $\frac{2}{5}$ of water, how many cm^3 of water is left in box B? (3 points)

Tips: (c) Water level height = volume of water $\div$ B bottom area of the box. Check whether the B box height is exceeded.

Answer area - please write all calculation steps and complete answer sentences:

— Complete the entire volume —

Total score: _____ / 100 points

Signature of marking teacher: _____ Date: _____

Exam tips:

- Part A MC Traps: Question 3: The Tian shape cannot be folded into a cube, Question 7: Decimal drainage method, Question 8: Radius vs diameter relationship, Question 9: Volume magnification
- Part B calculation questions: LCM must be found for general fractions, the division sign must be changed into a multiplication sign, and the divisor must be reversed when dividing a fraction, and the area of a trapezoid must be divided by 2
- Part C application questions: Each question contains cross-topic traps (unit conversion, water level height inversion, space residual calculation)
- Make good use of 50 minutes: 12 minutes for Part A $\rightarrow$ 18 minutes for Part B $\rightarrow$ 16 minutes for Part C $\rightarrow$ 4 minutes for inspection

Appendix: Simulation 3 Post-examination Supplementary Practice

24 questions · For after-class
consolidation

Instructions: The following questions are designed for the core concepts of Simulation 3 in the second semester (circle, volume, multiplication and division of fractions).

Understanding circle (total 5 questions)

#	Question	Difficulty
S1	The diameter of a circle is 50 cm. radius = ? $\pi \approx 3.14$, the circumference of a circle is approximately = ?	
S2	The radius of a circle is 7 cm. Diameter = ? Circle area = ? ($\pi \approx 22/7$)	
S3	The circumference of the circular garden bed is approximately 62.8 m. What is its approximate diameter? ($\pi \approx 3.14$)	
S4	Which of the following statements about circles is true? (a) The diameter is 2 times the radius (b) The radius is 2 times the diameter (c) All diameters are equal in length (d) The longer the radius, the smaller the area	
S5	The diameter of a circle is 24 cm. Cut out the largest square from this circle (the vertex of the square is on the circumference), the area of the square = ?	

3D shape and volume (total 6 question)

#	Question	Difficulty
S6	The cuboid is 9 cm long, 4 cm wide, and 6 cm high. Volume = ? Surface area = ?	

S7	The side length of the cube is 5 cm. Volume = ?	
S8	The side length of the cube water tank is 25 cm. How many liters of water can it hold? (1 L = 1000 cm ³)	
S9	The length, width and height of a cuboid are each magnified 3 times. How many times the new volume is the original?	
S10	Cuboid water tank A (40 × 30 × 20 cm) is filled with water and poured into cube water tank B (side length 30 cm). Will the water overflow? How high is the water level?	
S11	The side length of the cube is $2\frac{1}{2}$ cm. Volume = ? (Answer with mixed fractions)	

fractionmultiplydivision (total 6 question)

#	Question	Difficulty
S12	$\frac{2}{7} \times \frac{3}{5} = ?$	
S13	$\frac{5}{8} \div \frac{3}{4} = ?$	
S14	$2\frac{1}{3} \times 1\frac{2}{5} = ?$	
S15	$3\frac{1}{2} \div \frac{7}{10} = ?$	
S16	A box of sugar contains 120 pieces. Xiao Ming ate $\frac{1}{6}$, Xiaohua ate $\frac{2}{5}$. Who eats more? How many more pills? $\frac{1}{6}$, Xiaohua ate $\frac{2}{5}$. Who eats more? How many more pills?	
S17	SEP for a number is equal to 36. What is the $\frac{3}{4}$ of this number? $\frac{2}{3}$ How much?	

Cross-Topiccomprehensive (total 7 question)

#	Question	Difficulty
S18	The area of the base of a trapezoid (upper base 4 cm, lower base 8 cm, height 5 cm). If the cylinder with the base of this trapezoid is 10 cm high, its volume = ?	

S19	The interior of the rectangular water tank is $50 \times 40 \times 30$ cm. Contains $\frac{2}{5}$ water. How many liters of water is there?	
S20	The side length of cube water tank A is 40 cm, and the length of rectangular water tank B is 50 cm, width 30 cm, and height 25 cm. Fill A with water and pour it into B. How high is the water level in tank B?	
S21	The radius of a circle is 14 cm. ($\pi \approx 22/7$) Find (a) the circumference of the circle (b) the area of the circle.	
S22	Cube expansion diagram: Which expansion diagram cannot be folded into a cube? Please give an example.	
S23	The length of a rectangle increases by 3 cm and its width decreases by 2 cm. Original length 12 cm, width 8 cm. How much has the area changed?	
S24	A cube water tank has a side length of 30 cm and contains $\frac{2}{3}$ water. If a stone with a volume of 5000 cm^3 is thrown in, will the water overflow?	

 Post-exam review record:

Part A score: ____/30 Part B score: ____/40 Part C score: ____/30 Total score: ____/100

The question types that need to be strengthened most in this exam: _____; The traps that need to be reviewed: _____

